

COMPUTER FUNDAMENTALS

Unit-II: Memory and Secondary Storage Devices

Comprehensive Study Notes

Topic 1: Memory Hierarchy

1.1 What is Memory Hierarchy?

Memory hierarchy is the organized structure of different types of memory in a computer system based on **speed**, **cost**, and **capacity**. The concept is simple: faster memory is more expensive and has smaller capacity, while slower memory is cheaper and has larger capacity. The hierarchy ensures that the CPU gets data quickly while keeping costs reasonable.

REGISTERS (Fastest)
Inside CPU
Smallest Capacity
CACHE MEMORY (Very Fast)
Between CPU & RAM
Small Capacity
MAIN MEMORY - RAM (Fast)
Moderate Capacity
Volatile
SECONDARY STORAGE (Slow)
Large Capacity
Non-Volatile
TERTIARY STORAGE (Slowest)
Largest Capacity
Archival

1.2 The Five Levels of Memory Hierarchy

Level 1: Register Memory

Registers are the **fastest** memory in a computer system. They are small, high-speed storage locations located **inside the CPU**.

Feature	Description
Speed	Fastest memory type
Capacity	Extremely small (32-64 registers)
Cost	Most expensive memory
Access Time	Less than 1 nanosecond
Purpose	Stores immediate data and instructions

Types of Registers:

Register	Full Form	Function
AC	Accumulator	Stores results of operations
PC	Program Counter	Holds address of next instruction
IR	Instruction Register	Stores current instruction
MAR	Memory Address Register	Holds memory address
MDR	Memory Data Register	Holds data being transferred

Level 2: Cache Memory

Cache memory is a small, very fast memory located **between the CPU and RAM**.

Feature	Description
Speed	Very fast (faster than RAM)
Capacity	Small (1MB to 32MB)
Technology	Uses SRAM technology

Access Time	1-10 nanoseconds
Purpose	Stores frequently accessed data

How Cache Works:

1. CPU first checks cache for data (cache hit)
2. If found, CPU accesses quickly
3. If not found (cache miss), CPU gets data from RAM
4. Cache controller manages data flow

Types of Cache:

Type	Location	Size	Speed
L1 Cache	Inside CPU	32-64 KB	Fastest
L2 Cache	Between CPU & RAM	256-512 KB	Medium
L3 Cache	Shared among cores	8-32 MB	Slowest cache

Level 3: Main Memory (RAM)

Main memory (RAM) is the **primary storage** where data and programs are stored during processing.

Feature	Description
Speed	Moderate speed
Capacity	Large (4GB to 128GB)
Technology	Uses DRAM technology
Access Time	50-100 nanoseconds
Volatility	Volatile - Data lost when power is off

Level 4: Secondary Storage

Secondary storage provides **permanent data storage** with large capacity.

Feature	Description
Speed	Slow speed
Capacity	Large (500GB to 20TB)
Cost	Low cost
Volatility	Non-volatile - Data retained without power
Examples	HDD, SSD, CD, DVD

Level 5: Tertiary Storage

Tertiary storage is used for **archival purposes**.

Feature	Description
Speed	Slowest speed
Capacity	Largest capacity
Cost	Lowest cost
Purpose	Used for backup and archiving
Examples	Magnetic Tape, Optical Disks

1.3 Comparison Table

Memory Type	Speed	Capacity	Cost	Volatile?
Registers	Fastest	Smallest	Highest	Yes
Cache Memory	Very Fast	Small	Very High	Yes
RAM	Fast	Moderate	High	Yes
Hard Disk	Slow	Large	Low	No

Tape	Slowest	Largest	Lowest	No
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Topic 2: Main Memory Organization

2.1 Memory Measurement Units

Unit	Value
Bit	Binary digit (0 or 1)
Byte	8 bits
KB	Kilobyte = 1024 bytes
MB	Megabyte = 1024 KB
GB	Gigabyte = 1024 MB
TB	Terabyte = 1024 GB
PB	Petabyte = 1024 TB

2.2 Memory Organization

Main memory is organized as a collection of **memory cells**, each with a unique **address**. This allows the CPU to directly access any memory location.

- **Memory Address:** A unique number that identifies each memory location
- **Word:** The amount of data accessed in one memory operation (typically 8, 16, 32, or 64 bits)

Topic 3: RAM (Random Access Memory)

3.1 Definition

RAM (Random Access Memory) is the **main memory** where data and programs are temporarily stored for processing.

3.2 Characteristics of RAM

Feature	Description
Volatility	Data lost when power is off
Access	Read and Write operations allowed
Speed	Fast access
Purpose	Running programs and processing data

3.3 Types of RAM

SRAM (Static RAM)

SRAM uses flip-flop circuits to store each bit of data. It is called "static" because it does not need to be refreshed.

Feature	Description
Technology	Uses flip-flops to store data
Speed	Faster than DRAM
Cost	More expensive
Refreshing	Not needed

Usage	Used as Cache Memory
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Advantages:

- Very fast speed
- No refreshing needed
- Low power consumption

Disadvantages:

- Expensive
- Lower density (less capacity per chip)

DRAM (Dynamic RAM)

DRAM uses capacitors to store each bit of data. It is called "dynamic" because it needs to be refreshed periodically.

Feature	Description
Technology	Uses capacitors to store data
Speed	Slower than SRAM
Cost	Cheaper than SRAM
Refreshing	Needed periodically
Usage	Used as Main Memory

Advantages:

- Cheap
- High density (more capacity per chip)

Disadvantages:

- Slower than SRAM
- Needs refreshing
- Higher power consumption

3.4 SRAM vs DRAM Comparison

Feature	SRAM	DRAM
Speed	Faster	Slower
Cost	Expensive	Cheap
Capacity	Small	Large
Refreshing	Not needed	Needed
Power	Low	High
Used as	Cache Memory	Main Memory

Topic 4: ROM (Read Only Memory)

4.1 Definition

ROM (Read Only Memory) is **non-volatile** memory that stores **permanent** data.

4.2 Characteristics of ROM

Feature	Description
Non-volatile	Data retained without power
Read Only	Data cannot be modified easily
Purpose	Storing firmware and boot programs

4.3 Types of ROM

MROM (Masked ROM)

Programmed during manufacturing. Cannot be changed.

- Programmed at factory
- Cannot be altered
- Used for permanent data
- Cheapest type of ROM

PROM (Programmable ROM)

Can be programmed once by the user using a special device called a PROM programmer.

- Programmed once by user
- Cannot be erased
- One-time programmable

EPROM (Erasable PROM)

Can be erased using **UV light** and reprogrammed.

- Erased using UV light
- Can be reprogrammed
- Takes 15-20 minutes to erase
- Has a quartz window for UV exposure

EEPROM (Electrically Erasable PROM)

Can be erased using **electrical signals** and reprogrammed.

- Erased electrically
- Fast erasing (milliseconds)
- Can be reprogrammed many times
- Used in modern devices

4.4 ROM Types Comparison

Type	Programmable	Erasable	Method	Speed
MROM	No	No	N/A	Fast
PROM	Once	No	N/A	Fast
EPROM	Multiple	Yes	UV Light	Moderate
EEPROM	Multiple	Yes	Electrical	Slower

Topic 5: Cache Memory

5.1 Definition

Cache memory is a **small, fast memory** placed between the **CPU and RAM** to improve processing speed.

5.2 Characteristics of Cache Memory

- Faster than RAM
- Smaller capacity
- Stores frequently accessed data
- Improves processing speed
- Uses SRAM technology

5.3 Why Cache Memory is Used?

Problem: CPU is much faster than RAM.

Solution: Cache memory bridges this speed gap.

How It Works:

1. Frequently used data is stored in cache
2. CPU first checks cache (cache hit)
3. If found, CPU accesses data quickly
4. If not found (cache miss), CPU gets data from RAM

5.4 Types of Cache Memory

Type	Location	Size	Speed
L1 Cache	Inside CPU	32-64 KB	Fastest
L2 Cache	Between CPU & RAM	256-512 KB	Medium

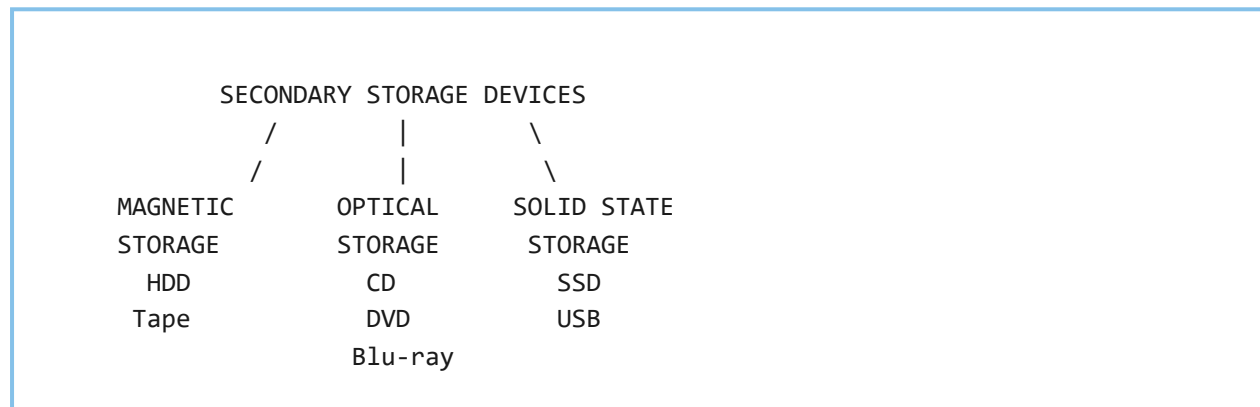
L3 Cache	Shared among cores	8-32 MB	Slowest
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Topic 6: Secondary Storage

6.1 What is Secondary Storage?

Secondary storage provides **permanent data storage**. It is **non-volatile** and has **large capacity**.

6.2 Classification of Secondary Storage



Topic 7: Magnetic Storage

7.1 Magnetic Tape

One of the oldest storage media. Data is stored magnetically on a plastic tape.

Feature	Description
Access	Sequential access only
Capacity	High storage capacity
Speed	Slow access speed
Usage	Backup and archiving
Cost	Low cost per GB

Advantages:

- Large storage capacity
- Low cost
- Durable
- Long shelf life

Disadvantages:

- Sequential access only
- Slow speed
- Not suitable for frequent access

7.2 Internal Hard Disk (HDD)

The primary storage device in most computers. Stores data on rotating magnetic disks.

How It Works:

1. Platters (disks) rotate at high speed (5400-7200 RPM)
2. Read/Write head moves across platters
3. Data stored magnetically on platters
4. Platters divided into tracks and sectors

Feature	Description
Access	Direct access storage
Capacity	Large (500GB to 20TB)
Speed	Moderate
Volatility	Non-volatile

Advantages:

- High storage capacity
- Low cost per GB
- Reliable
- Good for large files

Disadvantages:

- Moving parts can fail
- Slower than SSD
- Fragile to physical shock
- Consumes more power
- Generates heat and noise

7.3 External Hard Drives

Type	Description
Portable HDD	Small, powered by USB
Desktop HDD	Larger, needs external power
NAS	Network attached storage

Topic 8: Optical Storage

8.1 What are Optical Storage Devices?

Devices that use **laser technology** to read and write data. Data is stored on reflective surfaces.

8.2 CD (Compact Disk)

Optical disk used for storing music, software, and data.

Feature	Description
Capacity	650-700 MB
Technology	Uses red laser

Types of CDs:

Type	Description
CD-ROM	Read only
CD-R	Write once
CD-RW	Rewritable

8.3 DVD (Digital Versatile Disk)

High-capacity optical disk used for movies, software, and data.

Feature	Description
Capacity	4.7 GB (single layer), 8.5 GB (dual layer)

Technology	Uses red laser
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Types of DVDs:

Type	Description
DVD-ROM	Read only
DVD-R	Write once
DVD-RW	Rewritable
DVD+R	Alternative write once
DVD+RW	Alternative rewritable

DVD vs CD Comparison:

Feature	CD	DVD
Capacity	700 MB	4.7 GB
Use	Music	Movies
Quality	Lower	Higher

8.4 Blu-ray Disk

High-definition optical disk.

Feature	Description
Capacity	25 GB (single layer), 50 GB (dual layer)
Technology	Uses blue laser
Usage	HD movies

Advantages:

- Much higher capacity
- Higher quality
- More durable

Topic 9: Solid State Storage

9.1 Flash Memory

Non-volatile memory that uses no moving parts.

Feature	Description
Technology	No moving parts
Speed	Fast access speed
Durability	Durable
Power	Low power consumption

Advantages:

- Very fast
- Durable
- Low power
- Silent

Disadvantages:

- More expensive per GB
- Limited write cycles

9.2 USB Drives

Portable flash memory devices.

Feature	Description
Portability	Very portable
Ease	Plug and play

Capacity	2GB to 1TB
Power	No external power needed

Uses:

- File transfer
- Data backup
- Bootable drives

9.3 Solid State Drive (SSD)

Modern storage device using flash memory.

Feature	Description
Speed	Faster than HDD (500 MB/s)
Cost	More expensive than HDD
Moving Parts	No moving parts
Usage	Used in modern laptops and PCs

Advantages over HDD:

- Much faster (10x faster)
- No noise
- Lower power consumption
- More durable
- Lighter weight

Disadvantages:

- Higher cost
- Limited capacity
- Limited write cycles

9.4 SSD vs HDD Comparison

Feature	HDD	SSD
Speed	Slower (100 MB/s)	Faster (500 MB/s)
Cost	Lower	Higher
Durability	Less	More
Noise	Yes	No
Power	More	Less
Weight	Heavier	Lighter
Capacity	Larger (20TB)	Smaller (2TB)

Topic 10: Comparison Tables

10.1 Comparison of Storage Devices

Device	Access	Speed	Capacity	Cost
HDD	Direct	Moderate	Large	Low
SSD	Direct	Fast	Medium	High
CD	Direct	Slow	Small	Low
DVD	Direct	Slow	Medium	Low
Blu-ray	Direct	Slow	Large	Medium
USB	Direct	Fast	Medium	Medium
Tape	Sequential	Very Slow	Very Large	Very Low

10.2 Primary vs Secondary Memory

Feature	Primary Memory	Secondary Memory
Volatility	Volatile	Non-volatile
Speed	Fast	Slow
Capacity	Small	Large
Cost	High	Low
Access	Direct	Direct/Sequential

Topic 11: Quick Revision

11.1 Memory Hierarchy Summary

Registers --> Cache Memory --> RAM --> Hard Disk --> Tape
(Fast) (Slow)

11.2 Memory Types Summary

Type	Volatile	Purpose
RAM	Yes	Temporary storage
ROM	No	Permanent storage
Cache	Yes	Speed improvement

11.3 Storage Types Summary

Type	Examples
Magnetic	HDD, Tape
Optical	CD, DVD, Blu-ray
Solid State	SSD, USB

Questions

Short Answer Questions (2-5 Marks)

1. What is memory hierarchy?
2. Explain the difference between SRAM and DRAM.
3. What is cache memory? Why is it used?
4. What is the difference between RAM and ROM?
5. Explain PROM, EPROM, and EEPROM.
6. What is secondary storage? Why is it needed?
7. Compare magnetic and optical storage.
8. What is the difference between CD, DVD, and Blu-ray?
9. What is a USB drive? What are its advantages?
10. What are the types of cache memory?

Long Answer Questions (10 Marks)

1. Explain the memory hierarchy with a neat diagram.
2. Describe different types of ROM - PROM, EPROM, EEPROM.
3. Discuss various secondary storage devices. Compare magnetic, optical, and solid-state storage.
4. Describe cache memory - its need, function, and types.
5. Explain different types of optical storage devices.
6. Explain RAM and ROM in detail with their types.
7. Describe magnetic tape storage - advantages and limitations.
8. Explain the working of a hard disk drive.
9. Compare SSD and HDD in detail.

Objective Questions (1 Mark)

1. Which memory is used as cache? (a) DRAM (b) SRAM (c) ROM (d) HDD

Answer: (b)

2. Which is volatile memory? (a) ROM (b) RAM (c) HDD (d) DVD

Answer: (b)

3. Which memory can be erased using UV light? (a) PROM (b) EPROM (c) EEPROM (d) MROM

Answer: (b)

4. The fastest memory is: (a) RAM (b) Cache (c) Register (d) HDD

Answer: (c)

5. Magnetic tape uses: (a) Direct access (b) Sequential access (c) Random access (d) None

Answer: (b)

6. CD capacity is: (a) 4.7 GB (b) 700 MB (c) 25 GB (d) 1 GB

Answer: (b)

7. DVD capacity is: (a) 700 MB (b) 4.7 GB (c) 25 GB (d) 1 TB

Answer: (b)

End of Unit-II Notes